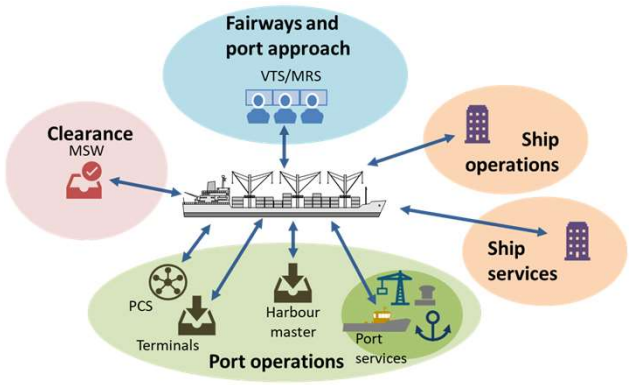
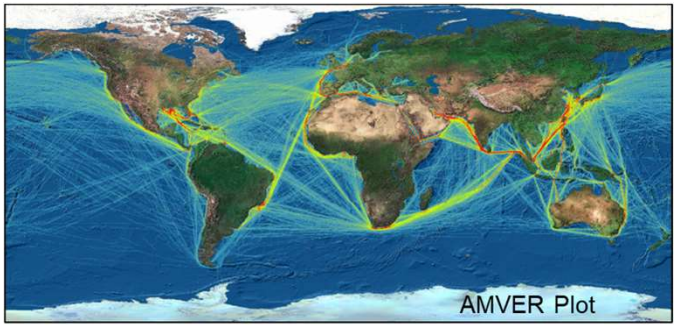
A brief overview of ISO 28005

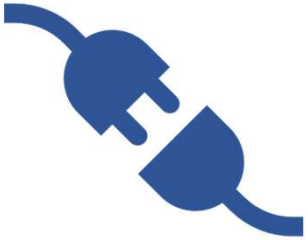
Purpose of ISO 28005



Ship-centric ship-shore operational and administrative communication.

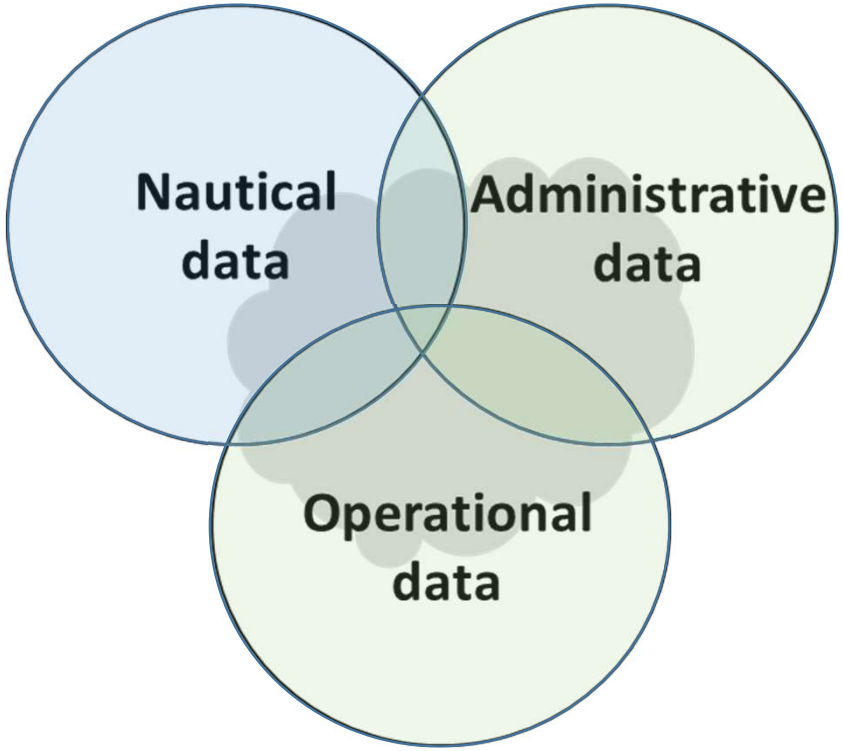


Facilitate communication between all ships and shore in all ports and countries.



Technical specification enabling consistent development of APIs.

ISO 28005 conforms to the IMO Compendium

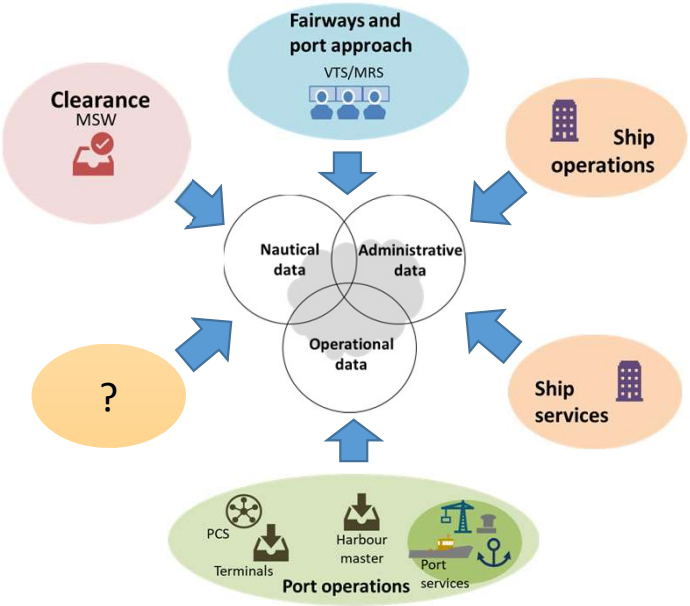


https://imo.org/en/OurWork/Facilitation/Pages/IMOCompendium.aspx

International collaborative effort







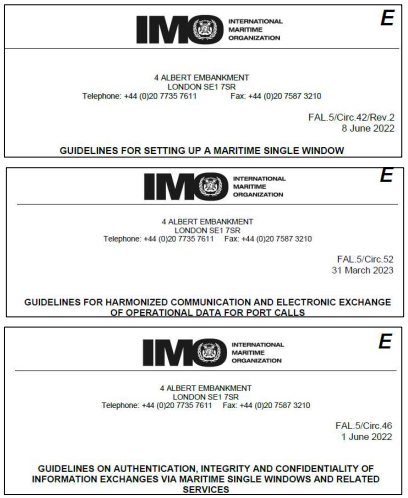
2000: The FAL Compendium defined how the FAL Convention could be implemented by electronic means.

2019: Establishment of the new IMO Compendium

>2021: IMO Compendium becomes a focal point for further international standardization of ship-shore cooperative processes.

ISO 28005 series provides technical specifications for the maritime community

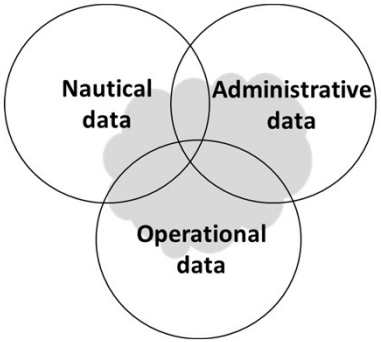
Main components of ISO 28005



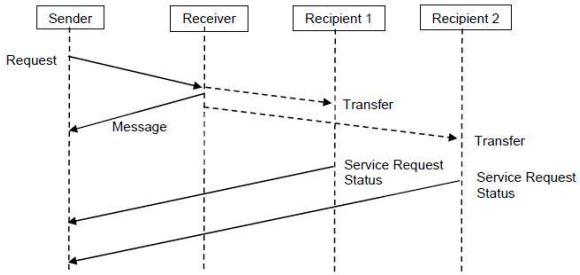
Principles and semantics for data exchanges between ship and shore in line with IMO guidelines.



A technical specification for development of a set of consistent APIs to support a business process.



A data model where data elements can be selected to meet the requirements of the business process.

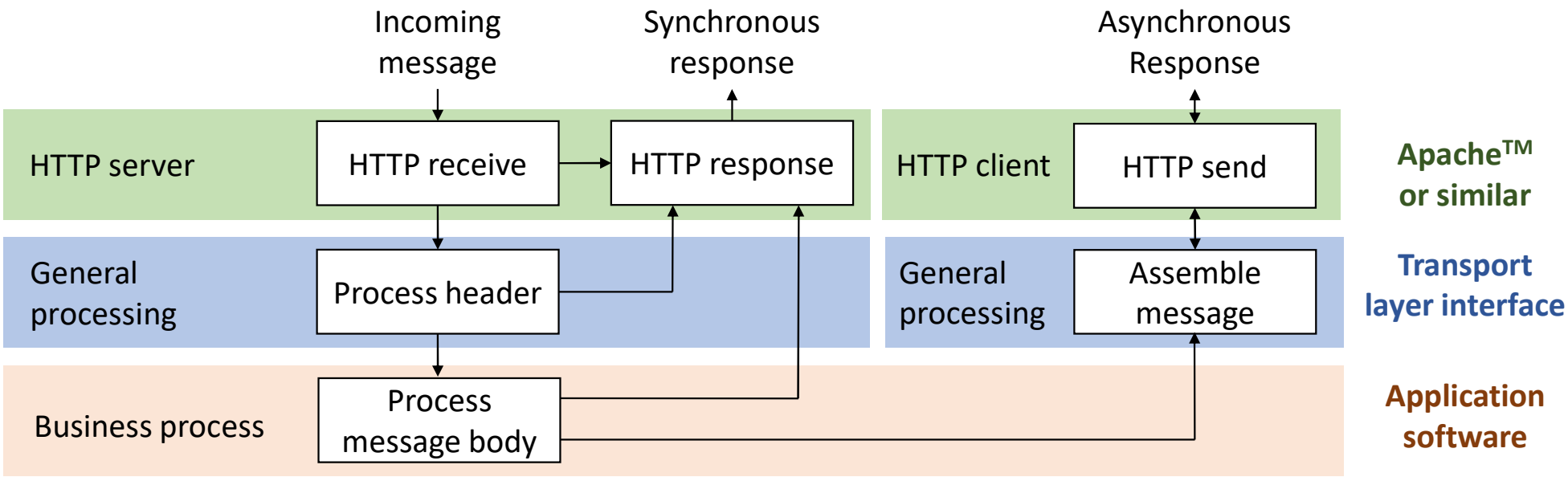


Message Implementation Guides (MIG) that specify the APIs to support a business process, e.g. maritime single window, just in time etc.

ISO 28005-1 (API design principles)

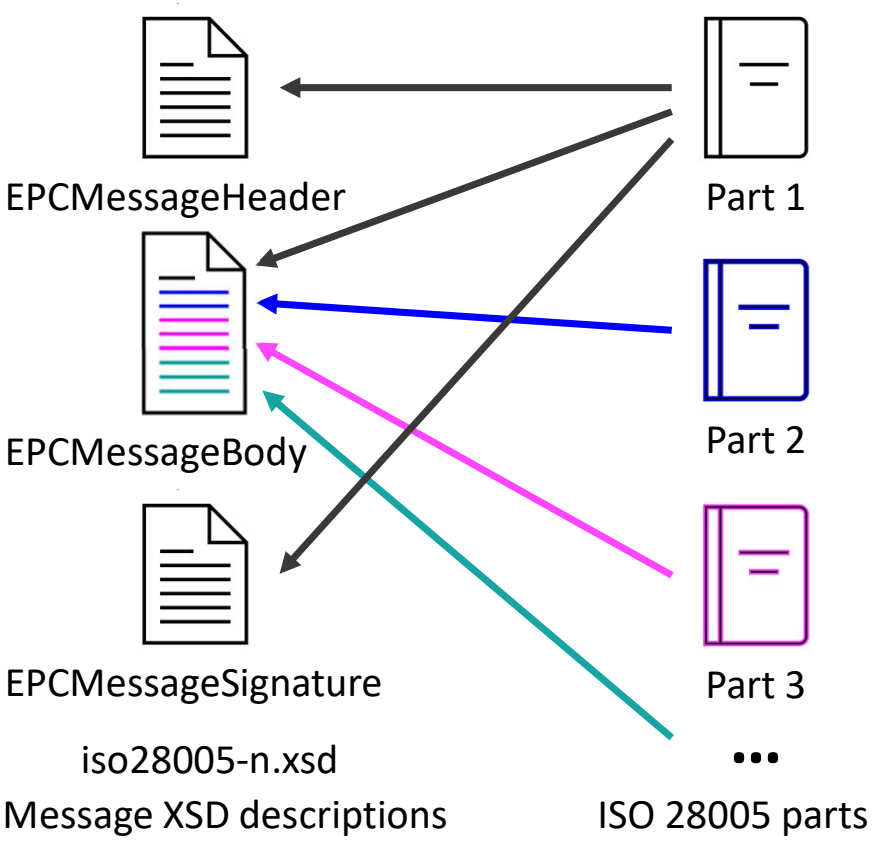
ISO 28005-2,3,… (business processes and data elements)

Technical specification for API implementation



ISO 28005-1

Extensible multi-part API message structure



Message header (XML)		Message header Always one
Sender identity Message reference Acknowledgement method Sent time	Receiver identity Service name Service request reference Service request type …	
Message body (XML, JSON, EDIFACT …)		Add for data content Zero or one
Data required to implement the requested service as defined in other parts of ISO 28005.		
Attachments (PDF, PNG, JPG, XML …)		Attachments Zero or more
Attachments referenced in the message body or header		
X.509 certificates (PEM)		Public key certificates Zero or more
X.509 certificate for signatories		
Digital signature (XML)		Digital signatures Zero or one
Authentication, Integrity, Non-repudiation …		

Current Status

- ISO 28005-1: DIS, Voting closes December 8. 2023.
- ISO 28005-2: Published 2021 – some amendments in new 28005-1/3.
- ISO 28005-3: Release as DIS in October for voting.

- New amendment procedure from 2024. To be announced.

- Overall goals:
  - Always mapped to IMO Compendium
  - Publish consolidated XSD files when new parts are released
  - May also publish EA-model (but need to maintain consistency)